Project 1 in FYS5419 – Quantum computing and quantum machine learning

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We have all the code needed to produce the plots and solve problems numerically in our GitHub Folder [1].

Exercises

Problem (a)

We fix a one-qubit basis, where we have chosen the standard computational basis

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad , \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

We have the Pauli matrices, given by

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad , \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad , \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

with the following actions on the basis

$$\begin{aligned} X|0\rangle &= |1\rangle, & X|1\rangle &= |0\rangle, \\ Y|0\rangle &= i|1\rangle, & Y|1\rangle &= -i|0\rangle, \\ Z|0\rangle &= |0\rangle, & Z|1\rangle &= -|1\rangle. \end{aligned}$$

It is worth noting that

$$ZX = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

$$-i\,ZX = -i \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = Y.$$

Moreover, the Hadamard gate is given by

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

while the pahse gate is given by

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix},$$

with the following basis actions

$$\begin{aligned} H|0\rangle &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} := |+\rangle, & H|1\rangle &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} := |-\rangle, \\ S|0\rangle &= |0\rangle, & S|1\rangle &= i|1\rangle. \end{aligned}$$

As can be seen if one runs the p1.ipynb file in the GitHub we get the results given in by applying the all the aforementioned gates to the computational basis states.

Gate	**Gate($\|0\rangle$)**	**Gate($\|1\rangle$)**
I	[1.0, 0.0]	[0.0, 1.0]
Z	[1.0, 0.0]	[0.0, -1.0]
X	[0.0, 1.0]	[1.0, 0.0]
Y	[0, i]	[$-i$, 0]
H	[0.707, 0.707]	[0.707, -0.707]
Phase	[1, 0]	[0, i]

TABLE I: Action of single-qubit gates on computational basis states.

As expected the identity leaves everything unchanged. Z applies a phase factor of -1 to $|1\rangle$ and acts trivially on $|0\rangle$. X acts as a flip between $|0\rangle$ and $|1\rangle$. Applying Y first flips the basis states and then adds a phase of i and $-i$ to $|0\rangle$ and $|1\rangle$, respectively. This makes sense since $Y = -iZX$. Applying H creates a superposition with amplitude $\frac{1}{\sqrt{2}} \approx 0.707$ for both basis states when applied to $|0\rangle$, and a superposition with amplitudes $\frac{1}{\sqrt{2}}$ and $-\frac{1}{\sqrt{2}}$ for $|0\rangle$ and $|1\rangle$, respectively, when applied to $|1\rangle$. The S-gate acts trivially on $|0\rangle$ and adds a complex phase of i when applied to $|1\rangle$. We see that the analytical and numerical results match.

For a two-qubit system, we can define a computational basis by $|ij\rangle := |i\rangle \otimes |j\rangle$ for $i, j = 0, 1$. This allows us to define the maximally entangled bell states, given by

$$\left|\Phi^+\right\rangle := \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad \left|\Phi^-\right\rangle := \frac{|00\rangle - |11\rangle}{\sqrt{2}},$$
$$\left|\Psi^+\right\rangle := \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad \left|\Psi^-\right\rangle := \frac{|01\rangle - |10\rangle}{\sqrt{2}}.$$

A useful 2-qubit gate is the $CNOT$ gate, given by

$$CNOT = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X,$$

which corresponds to the mapping

$$|ij\rangle \mapsto |i, i \oplus j\rangle\,,$$

where $\oplus$ denotes the XOR operation. Note that applying the Hadamard gate on a single qubit, followed by the $CNOT$ gate on $|00\rangle$ gives us

$$CNOT(H \otimes I)\,|00\rangle = \left|\Phi^+\right\rangle.$$

We define a new state $|\psi\rangle$ by applying the same gates again, which gives us

$$\begin{aligned}
|\psi\rangle &:= CNOT(H \otimes I)\left|\Phi^+\right\rangle \\
&= \frac{1}{\sqrt{2}} CNOT(H \otimes I)(|00\rangle + |11\rangle) \\
&= \frac{1}{\sqrt{2}} CNOT(|{+0}\rangle + |{-1}\rangle) \\
&= \frac{1}{2} CNOT(|00\rangle + |10\rangle + |01\rangle - |11\rangle) \\
&= \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle + |11\rangle) \\
&= \sqrt{2}\left(\left|\Phi^+\right\rangle + \left|\Psi^-\right\rangle\right).
\end{aligned}$$

In our code, we measure each single qubit several times, and empirically extract information on the states, by looking at averages of the measurements. In practice, our measurement probabilities are given by,

$$p_{ij} = |\langle ij|\psi\rangle|^2.$$

In the code we first measure measure the qubit in system A, where the probabilities are given by

$$p_A(0) = |\langle 00|\psi\rangle|^2 + |\langle 01|\psi\rangle|^2 = \frac{1}{2},$$

and

$$p_A(1) = |\langle 10|\psi\rangle|^2 + |\langle 11|\psi\rangle|^2 = \frac{1}{2}.$$

After measuring qubit A, our post measurement states become

$$|\psi\rangle_{A=0} = \frac{|00\rangle + |01\rangle}{\sqrt{2}},$$

and

$$|\psi\rangle_{A=1} = \frac{-|10\rangle + |11\rangle}{\sqrt{2}},$$

with corresponding probabilities

$$p_{B|A=0}(0) = \frac{1}{2} \quad p_{B|A=0}(1) = \frac{1}{2} \tag{1}$$

$$p_{B|A=1}(0) = \frac{1}{2} \quad p_{B|A=1}(1) = \frac{1}{2}. \tag{2}$$

Thus the joint probabilities are

$$p_{00} = p_{01} = p_{10} = p_{11} = \frac{1}{4}.$$

Correspondingly, the expectation values for system A and B are therefore given by

$$\mathbb{E}[A] = p_A(1) = \frac{1}{2},$$

and

$$\mathbb{E}[B] = p_B(1) = \frac{1}{2}.$$

For system A this gives us standard deviation

$$\sigma_A = (\mathbb{E}[A^2] - \mathbb{E}[A]^2)^{\frac{1}{2}} = (\mathbb{E}[A] - \mathbb{E}[A]^2)^{\frac{1}{2}} = \sqrt{\frac{1}{2} - \frac{1}{4}} = \frac{1}{2}.$$

Similarly for system B we get,

$$\sigma_B = \frac{1}{2}.$$

We will also look at the standard error, SE_i defined as $SE_i = \sigma_i/\sqrt{N}$, for $i = A, B$ and N is the number of shots. When $N \to \infty$ the numerator $\sigma_i \to 1/2$, and the denominator will go to ∞ so we expect SE_i will get closer to zero when increasing the number of shots.

In our code, we measure repeatedly, hence, our empirical mean will converge to $\frac{1}{2}$ for each qubit. In table II we see the results and corresponding statistics of our measurement procedure. We ran our simulations across different numbers of shots. The table includes the four different computational basis states, the count for each after performing the measurements, and the corresponding probabilities, as well as the mean, standard deviation and standard error for each qubit. Increasing the number of shots clearly increases accuracy. The mean and standard deviation for each qubit approaches $\frac{1}{2}$. The standard error correspondingly gets closer and closer to 0. We also see that the two-qubit basis states have a probability that goes to $\frac{1}{4}$ as expected. For the rest of the simulations we usually run with 10000 shots as this doesn't take too long to run, while also giving fairly accurate results.

We can also "trace out" a density matrix with respect to a subsystem. For $|\Phi^+\rangle$ in the computational basis with density matrix $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$, this corresponds to

$$\rho_A := \mathrm{Tr}_B(|\Phi^+\rangle\langle\Phi^+|) = \sum_i (I_A \otimes \langle i_B|)|\Phi^+\rangle\langle\Phi^+|(I_A \otimes |i_B\rangle) = \frac{1}{2}I.$$

TABLE II: Measurement results grouped by number of shots. Each row lists joint counts, joint probabilities and the summary statistics (Mean / Std / SE) for qubit 0 and qubit 1.

Shots	**Joint counts**	**Joint probabilities**	**Qubit 0 (Mean / Std / SE)**	**Qubit 1 (Mean / Std / SE)**
1000	00: 241 10: 246 01: 267 11: 246	00: 0.241000 10: 0.246000 01: 0.267000 11: 0.246000	Mean: 0.492000 Std: 0.499935996 SE: 0.015809364	Mean: 0.513000 Std: 0.499830971 SE: 0.015806043
10000	00: 2541 10: 2559 01: 2449 11: 2451	00: 0.254100 10: 0.255900 01: 0.244900 11: 0.245100	Mean: 0.501000 Std: 0.499999000 SE: 0.004999990	Mean: 0.490000 Std: 0.499899990 SE: 0.004999000
100000	00: 24764 10: 24993 01: 25099 11: 25144	00: 0.247640 10: 0.249930 01: 0.250990 11: 0.251440	Mean: 0.501370 Std: 0.499998123 SE: 0.001581133	Mean: 0.502430 Std: 0.499994095 SE: 0.001581120
1000000	00: 250047 10: 249292 01: 250855 11: 249806	00: 0.250047 10: 0.249292 01: 0.250855 11: 0.249806	Mean: 0.499098 Std: 0.499999186 SE: 0.000499999	Mean: 0.500661 Std: 0.499999563 SE: 0.000500000

Likewise

$$\rho_B = \frac{1}{2}I,$$

Which means that each qubit is in the maximally entangled state. Hence, we can easily compute the Von Neumann entropy

$$S(\rho_A) = -\sum_i \lambda_i \log_2 \lambda_i = -(-\frac{1}{2} - \frac{1}{2}) = 1,$$

where $\lambda_i = 1/2$ are the eigenvalues of ρ_A. Likewise

$$S(\rho_B) = 1,$$

which confirms that $|\Phi^+\rangle$ is maximally entangled. When running this numerically we similarily get a Von Neumann entropy of 1.

Problem (b)

As stated in the problem we can set

$$\begin{aligned} H(\lambda) &= H_0 + \lambda H_1 \\ &= \mathcal{E} I + \Omega\sigma_z + \lambda(cI + \omega_z\sigma_z + \omega_x\sigma_x) \\ &= (\mathcal{E} + \lambda c)I + (\Omega + \lambda\omega_z)\sigma_z + \lambda\omega_x\sigma_x. \end{aligned}$$

Meaning we get that

$$H(\lambda) = \begin{pmatrix} \mathcal{E} + \Omega + \lambda(c + \omega_z) & \lambda\omega_x \\ \lambda\omega_z & \mathcal{E} - \Omega + \lambda(c - \omega_z) \end{pmatrix}.$$

Thus, by quadratically solving the characteristic equation

$$\det(H(\lambda) - EI) = 0,$$

we get an expression for the eigenvalues as a function of λ, which is given by

$$E(\lambda) = (\mathcal{E} + \lambda c) \pm \sqrt{(\Omega + \lambda\omega_z)^2 + (\lambda\omega_z)^2}.$$

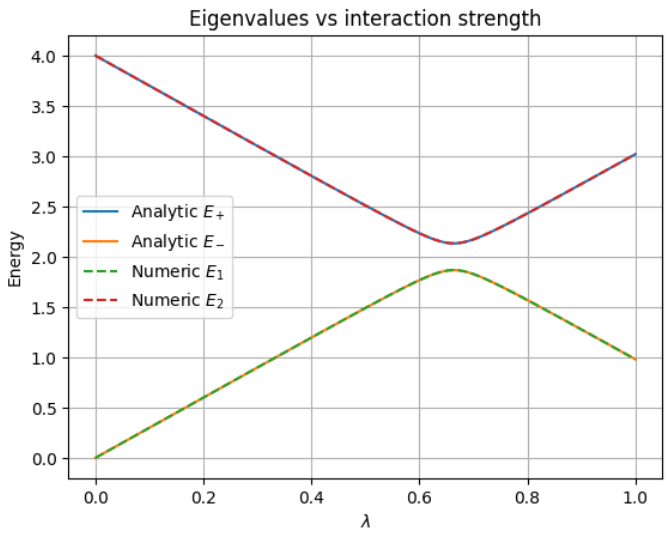


FIG. 1: We see that the analytical expression for the eigenvalues matches perfectly with the eigenvalues from the numerical solver across different values of λ. The avoided crossing near $\lambda \approx 2/3$ reflects the coupling between the basis states.

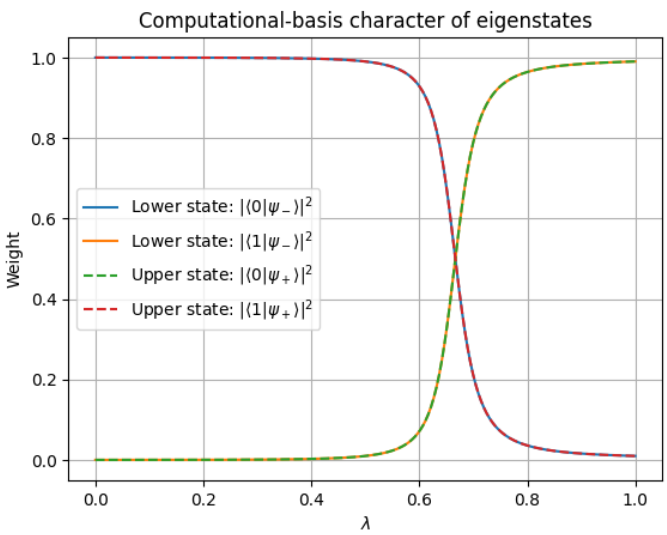


FIG. 2: The lower and upper eigenstates transition between $|0\rangle$ and $|1\rangle$, with an intersection at $\lambda \approx 2/3$.

We compare our analytical expression for the eigenvalues with the eigensolver from Numpy in figure 1. The avoided crossing near $\lambda \approx 2/3$ is a consequence of the coupling term, which mixes the basis states and prevents degeneracy. Moreover, we also look at the weights of the eigenstates as functions of the interaction strength in figure 2. We see that for small λ the lower eigenstate is dominated by $|0\rangle$, while the upper state is dominated by $|1\rangle$. As λ increases we get mixing between the basis states, and at $\lambda \approx 2/3$, we see that both $|0\rangle$ and $|1\rangle$ have approximately equal probability. For bigger λ we get an inverse of the small λ case.

Problem (c)

We first give a brief introduction to the variational quantum eigensolver (VQE), before we discuss the results of the implementation for our specific problem. Generally the VQE functions the following way. For a Hermitian H with ground state energy E_0, we know by the variational principle that

$$\langle\psi|H|\psi\rangle \geq E_0,$$

for any state $|\psi\rangle$, with equality if and only if $|\psi\rangle$ is the ground state. In VQE we use this fact and we introduce a parametrized trial state, given by

$$|\psi(\theta)\rangle := U(\theta)\,|0\rangle\,,$$

where $U(\theta)$ is a parametrized quantum circuit with initialized parameters θ. For our example, we parametrize the state by rotating angles on the Bloch sphere.

More specifically, in our implementation our circuit ansatz is

$$U(\theta) = R_z(\theta_3)R_y(\theta_2)R_z(\theta_1),$$

where R_x, R_y, R_z denote rotations about the Bloch sphere axes. When R_z acts on $|0\rangle$, the rotation only contributes a global phase, and therefore does not affect expectation values. We also tested a one parameter ansatz, given by

$$U(\theta) = R_y(\theta_0),$$

to compare with the performance of the three parameter ansatz. Hence, we get an optimization problem, where

$$E_0 \approx \min_\theta E(\theta), \quad E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle.$$

Since we can decompose our Hamiltonian into Pauli matrices, the problem is reformulated into minimizing expectation values of Pauli operators, which can be seen in our code. Meaning

$$E(\theta) = \sum_i c_i \langle\psi(\theta)|P_i|\psi(\theta)\rangle,$$

where c_i are the basis coefficients for our Hamiltonian, and P_i are the Pauli matrices. In our specific case, the Hamiltonian only contains the operators I, Z and X, and hence the energy reduces to a linear combination of the expectation values $\langle Z\rangle$ and $\langle X\rangle$.

We estimate the expectation values of the Pauli operators by repeated measurements, which gives an estimate for $E(\theta)$. In practice, we measure in the computational basis, corresponding to the Pauli matrix Z . Thus, since our Hamiltonian may contain the other Pauli matrices X and Y, we must rotate the basis before measuring. For example, if we have X, we do this by using the Hadamard gate to set

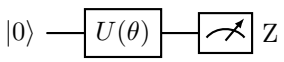
$$HXH = Z,$$

thus measuring X corresponds to measuring Z after applying the Hadamard gates. Since the Pauli matrices have eigenvalues ± 1, we can estimate the expectation values by

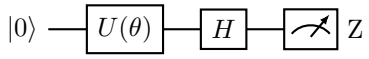
$$\langle\psi(\theta)|Z|\psi(\theta)\rangle \approx \frac{n_0 - n_1}{N},$$

where n_0 and n_1 are the number of outcomes corresponding to $|0\rangle$ and $|1\rangle$ respectively, and N is the number of shots. The circuits look as follows

Circuit for $\langle\sigma_z\rangle$:



Circuit for $\langle\sigma_x\rangle$:



Since the identity term is trivial we do not need a circuit for this term.

To optimize the parameters, we compute gradients using the parameter-shift rule. The derivatives can thus be written as

$$\frac{\partial E(\theta)}{\partial \theta_k} = \frac{E(\theta_k + \frac{\pi}{2}) - E(\theta_k - \frac{\pi}{2})}{2},$$

which is the method implemented in our code. Using Adam optimizer, we find optimal parameters θ, where the variational principle in theory guarantees that we do not undershoot, i.e. the estimate will always be an upper

Algorithm 1 Adam Optimizer

Initialize: $m_0 \leftarrow 0,\ v_0 \leftarrow 0,\ t \leftarrow 0$
Given: $\alpha,\ \beta_1,\ \beta_2,\ \epsilon$
while not converged **do**
  $t \leftarrow t + 1$
  $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$
  $m_t \leftarrow \beta_1 m_{t-1} + (1-\beta_1) g_t$
  $v_t \leftarrow \beta_2 v_{t-1} + (1-\beta_2) g_t^2$
  $\hat{m}_t \leftarrow \frac{m_t}{1-\beta_1^t}$
  $\hat{v}_t \leftarrow \frac{v_t}{1-\beta_2^t}$
  $\theta_t \leftarrow \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t}+\epsilon}$
end while

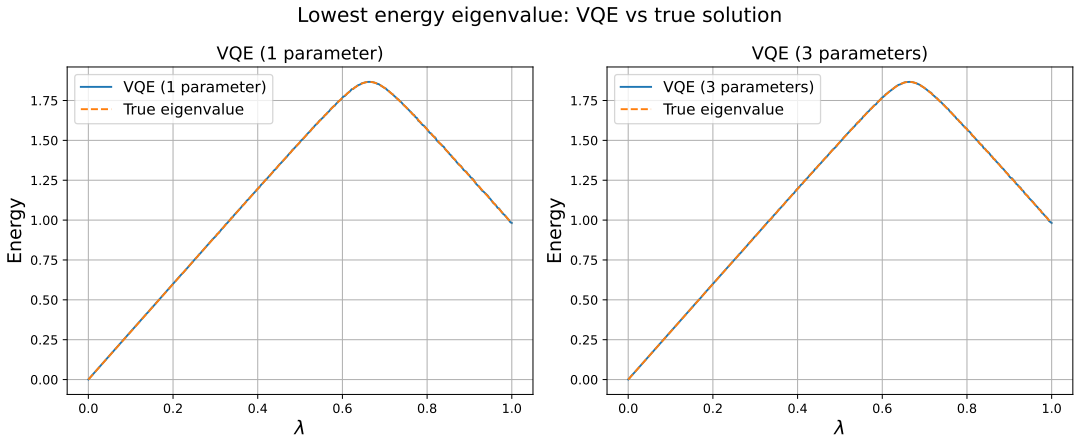


FIG. 3: Comparison between analytical solution for the ground state energy, found in problem B and VQE, with both the 1 parameter and 3 parameter ansatz. For 10 000 shots, the VQE overlaps almost perfectly with the analytical solution, for both the 1 parameter and 3 parameter ansatz.

bound of the true ground state energy (although small numerical fluctuations can sometimes violate this). The Adam optimizer is summarized in the algorithm 1. For more information on this optimizer see [2]. We have not experimented with different parameters, e.g. learning rate and the number of iteration steps, for the Adam optimizer as our results appear satisfactory.

To summarize, the VQE essentially rotates a qubit around the Bloch sphere until we find a minimum energy. In our results, we use the eigenvalues found in part (B) as a benchmark for our VQE routine with both the one parameter ansatz and the three parameter ansatz, and the comparison can be found in figure 3. We see that using 10000 shots is sufficient to capture the evolution of the ground state energy as a function λ. Moreover, the one parameter ansatz gives a more or less identical result to using the three parameter ansatz.

Moreover, in figure 4 we compare our VQE routine with the Qiskit VQE. This simulation was performed with 10000 shots, and we see that this yielded an almost exact overlap with the VQE from Qiskit.

Problem (d)

We now look at a two-qubit system. We again denote the first qubit as system A and the second as B. The computational basis states are

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix},\ |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix},\ |10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix},\ |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}.$$

Our Hamiltonian is $H = H_0 + \lambda H_I$, where H_0 is non-interacting and H_I is the interacting part. The parameter $\lambda \in [0,1]$ represents the interaction strength. H_0 is diagonal in the computational basis, defined by $H_0 = \mathrm{diag}(\epsilon_{00}, \epsilon_{10}, \epsilon_{10}, \epsilon_{11})$

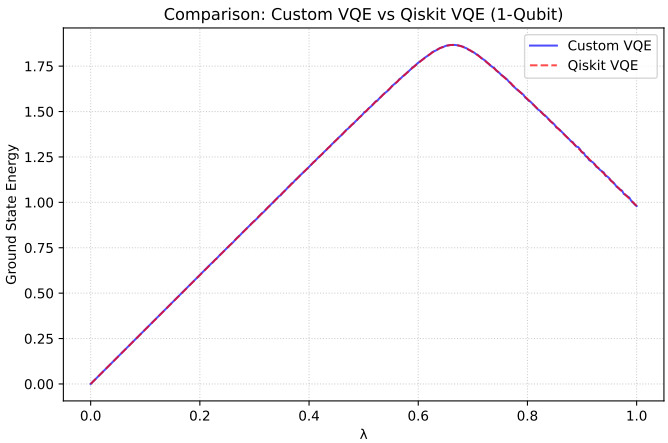


FIG. 4: Comparison between Qiskit and our implementation of VQE using ADAM as optimizer, with almost perfect overlap.

where the ϵ_{ij} are parameters. The interacting part of the Hamiltonian is given by $H_I = H_x X \otimes X + H_z Z \otimes Z$, for parameters H_x and H_z.
We also rewrite H_0 as a sum of Pauli matrices. Since H_0 is diagonal it can be written as a linear combination of tensor products of the identity I and the Pauli Z matrix.

$$H_0 = \alpha I \otimes I + \beta I \otimes Z + \gamma Z \otimes I + \delta Z \otimes Z.$$

Putting $\alpha = \frac{1}{4}(\epsilon_{00}+\epsilon_{01}+\epsilon_{10}+\epsilon_{11})$, $\beta = \frac{1}{4}(\epsilon_{00}+\epsilon_{01}-\epsilon_{10}-\epsilon_{11})$, $\gamma = \frac{1}{4}(\epsilon_{00}-\epsilon_{01}+\epsilon_{10}-\epsilon_{11})$, and $\delta = \frac{1}{4}(\epsilon_{00}-\epsilon_{01}-\epsilon_{10}+\epsilon_{11})$ yields the correct result. The tensor products are

$$I \otimes I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad I \otimes Z = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

$$Z \otimes I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad Z \otimes Z = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Now we compute each diagonal entry of H_0:

$$(H_0)_{11} = \alpha+\beta+\gamma+\delta = \frac{1}{4}(\epsilon_{00}+\epsilon_{01}+\epsilon_{10}+\epsilon_{11})+\frac{1}{4}(\epsilon_{00}+\epsilon_{01}-\epsilon_{10}-\epsilon_{11})+\frac{1}{4}(\epsilon_{00}-\epsilon_{01}+\epsilon_{10}-\epsilon_{11})+\frac{1}{4}(\epsilon_{00}-\epsilon_{01}-\epsilon_{10}+\epsilon_{11}) = \epsilon_{00},$$

$$(H_0)_{22} = \alpha-\beta+\gamma-\delta = \frac{1}{4}(\epsilon_{00}+\epsilon_{01}+\epsilon_{10}+\epsilon_{11})-\frac{1}{4}(\epsilon_{00}+\epsilon_{01}-\epsilon_{10}-\epsilon_{11})+\frac{1}{4}(\epsilon_{00}-\epsilon_{01}+\epsilon_{10}-\epsilon_{11})-\frac{1}{4}(\epsilon_{00}-\epsilon_{01}-\epsilon_{10}+\epsilon_{11}) = \epsilon_{01},$$

$$(H_0)_{33} = \alpha+\beta-\gamma-\delta = \frac{1}{4}(\epsilon_{00}+\epsilon_{01}+\epsilon_{10}+\epsilon_{11})+\frac{1}{4}(\epsilon_{00}+\epsilon_{01}-\epsilon_{10}-\epsilon_{11})-\frac{1}{4}(\epsilon_{00}-\epsilon_{01}+\epsilon_{10}-\epsilon_{11})-\frac{1}{4}(\epsilon_{00}-\epsilon_{01}-\epsilon_{10}+\epsilon_{11}) = \epsilon_{10},$$

$$(H_0)_{44} = \alpha-\beta-\gamma+\delta = \frac{1}{4}(\epsilon_{00}+\epsilon_{01}+\epsilon_{10}+\epsilon_{11})-\frac{1}{4}(\epsilon_{00}+\epsilon_{01}-\epsilon_{10}-\epsilon_{11})-\frac{1}{4}(\epsilon_{00}-\epsilon_{01}+\epsilon_{10}-\epsilon_{11})+\frac{1}{4}(\epsilon_{00}-\epsilon_{01}-\epsilon_{10}+\epsilon_{11}) = \epsilon_{11}.$$

Now we move on to actually computing the eigenvalues as a function of λ. This is done using the numpy.linalg.eigvalsh function found in the numpy library. We also discuss how the interaction strength λ affects entanglement. Thereafter we compute the Von Neumann entropy for the subsystem A using the density matrix

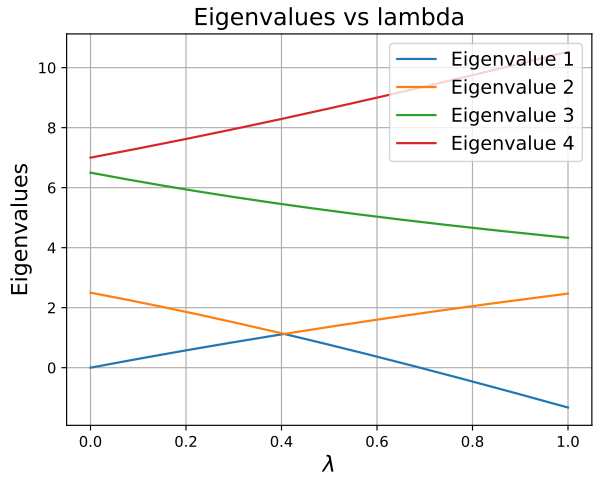


FIG. 5: Eigenvalues of Hamiltonian given in subproblem (4) as a function of the interaction strength λ.

of the state with the lowest energy.

The eigenvalues of H can be seen in Figure 5. At $\lambda \approx 0.4$ we observe degeneracy, the two lowest eigenvalues are equal. A level-crossing is happening at this point. This tells us that the lowest energy eigenstate changes at $\lambda = 0.4$. In Figures 6 and 7 we see how the squared norm of the coefficients of the computational basis states evolve as a function of λ for the lowest energy state. The lowest eigenstate starts as a pure state, $|00\rangle$. This can also be seen from the fact that when $\lambda = 0$, $H = H_0$, and H_0 is diagonal in the computational basis, giving a pure state. In our simulations ϵ_{00} was the lowest energy which implies the corresponding eigenstate is $|00\rangle$ as $(H_0)_{11} = \epsilon_{00}$.

Increasing λ yields mixing of basis states. For $\lambda < 0.4$ we see that the lowest state is a superposition of $|00\rangle$ and $|11\rangle$, where $|00\rangle$ is the dominant part. This state is an entangled state. This can be seen from the following: Define

$$|\psi\rangle = a|00\rangle + b|11\rangle.$$

An arbitrary product state can be written

$$(x|0\rangle + y|1\rangle) \otimes (u|0\rangle + v|1\rangle) = xu|00\rangle + xv|01\rangle + yu|10\rangle + yv|11\rangle.$$

Matching the terms requires $xv = 0, \quad yu = 0 \Rightarrow$ separable only if $a = 0$ or $b = 0$. In our case neither a, nor b is zero, so the state must be entangled. For $\lambda \in [0, 0.4)$ the weight of $|00\rangle$ decreases, while that of $|11\rangle$ increases, telling us that the states moves further away from a pure state and likely becomes more entangled.

At $\lambda \approx 0.4$, we observe a sudden transition in the set of basis states that make up the superposition. We go from the aforementioned $|00\rangle$ and $|11\rangle$ to $|01\rangle$ and $|10\rangle$. This is due to the level crossing that can be seen in Figure 5. After the crossing a different eigenstate has the lowest energy so the weights seen in 6 also change. This new state can be written as $|\phi\rangle = a_1 |01\rangle + a_2 |10\rangle$. Using the same argument as for $|\psi\rangle$ above, we see that $|\phi\rangle$ is entangled. $|\phi\rangle$ is not dominated as much as the previous lowest state by a single weight, i.e. the basis states are more mixed. This will lead to a sudden increase in the entanglement. For $\lambda > 0.4$ we see that the weights of $|\phi\rangle$ get closer in value. This tells us that we have even more mixing, and thus we expect entanglement to increase even more.

In 7 one can find the evolution of the weights for the second lowest eigenstate. Here we see that the state starts in a superposition of $|01\rangle$ and $|10\rangle$ where the basis states are more mixed than for the lowest state. After the crossing point it becomes a superposition of $|00\rangle$ and $|11\rangle$. The crossing of basis states here decreases and $|00\rangle$ dominates. This matches our discussion for the lowest energy state above.

The Von Neumann entropy calculated can be seen in figure 8. Here we see an increase in the entropy as λ grows, and a sudden spike close to the level-crossing at $\lambda = 0.4$. After this the entropy continues to increase. This confirms our discussion above.

It is worth to note that we would expect the Von Neumann Entropy of ρ_B to be identical since the eigenstates of the Hamiltonian are pure (regardless of the value of λ). For a pure bipartite system, the Von Neumann entropy of both reduced density matrices have the same Von Neumann entropy [3].

From the previous discussion, we can see that the interaction part of the Hamiltonian drives the entanglement, and that the strength of the interaction affects the degree of entanglement.

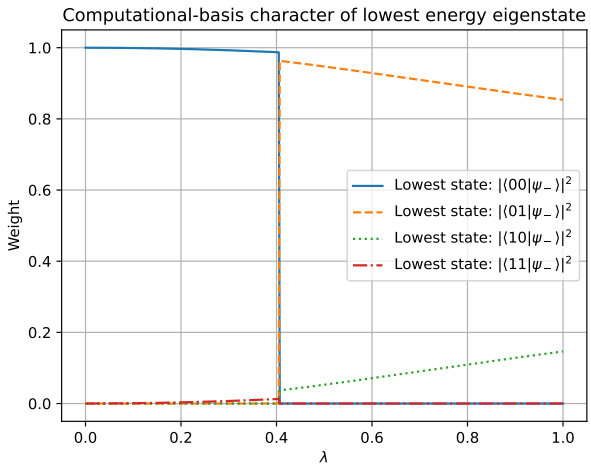


FIG. 6: Coefficients squared of the computational basis states for the eigenstate corresponding to the lowest energy as function of λ.

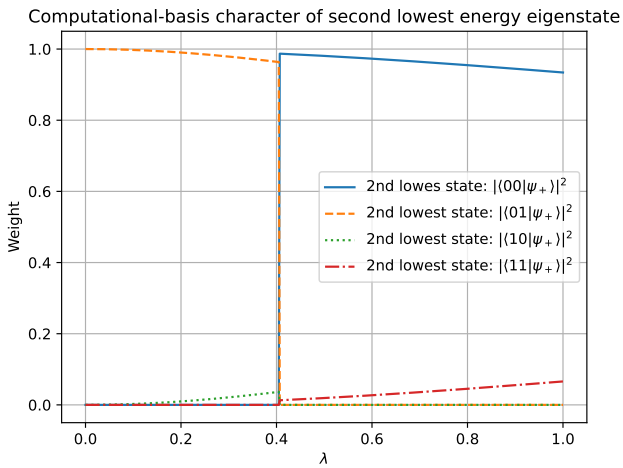


FIG. 7: Coefficients squared of the computational basis states for the eigenstate corresponding to the second lowest energy as function of λ.

Problem (e)

We extend the VQE method from part (c) to a two qubit state space. We prepare the trial state in the same fashion, with

$$|\psi(\theta)\rangle = U(\theta)\,|00\rangle\,.$$

In our implementation, the ansatz consists of two layers of single-qubit rotations with an entangling $CNOT$ gate in between. More precisely, we use

$$U(\theta) = U_2(\theta)\,CNOT\,U_1(\theta),$$

where $U_1(\theta)$ and $U_2(\theta)$ are tensor products of single-qubit rotation gates acting on each qubit separately. In our implementation specifically, we chose

$$U_1(\theta) = \big(R_x(\theta_0)\,R_z(\theta_1)\big)\ \otimes\ \big(R_x(\theta_2)\,R_z(\theta_3)\big),$$

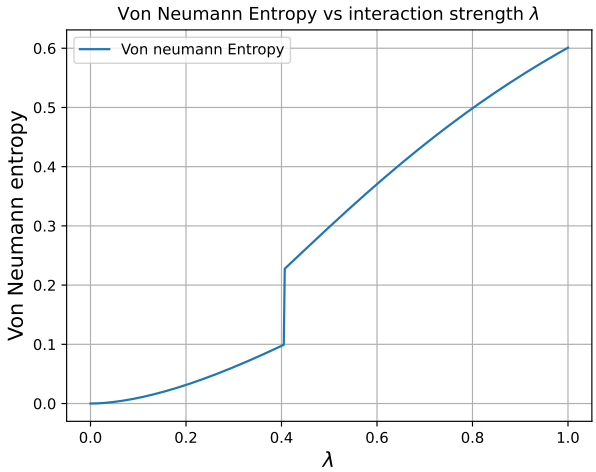


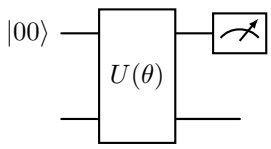
FIG. 8: Von Neumann Entropy of ρ_A as a function of the interaction strength λ.

and

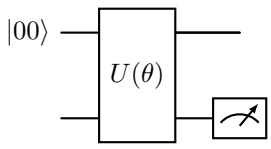
$$U_2(\theta) = \big(R_z(\theta_4)\, R_x(\theta_5)\big) \;\otimes\; \big(R_z(\theta_6)\, R_x(\theta_7)\big).$$

This structure allows the ansatz to represent both product states and entangled states. A schematic for the circuit can be written as follows, where we omit measures on the identity.

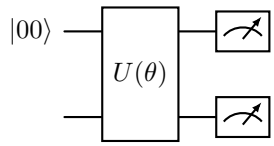
Circuit for $\langle Z \otimes I \rangle$:



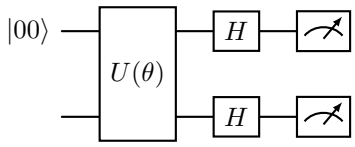
Circuit for $\langle I \otimes Z \rangle$:



Circuit for $\langle Z \otimes Z \rangle$:



Circuit for $\langle X \otimes X \rangle$:



From part (d) we have that

$$H = \alpha I \otimes I + \beta I \otimes Z + \gamma Z \otimes I + (\delta + \lambda H_z) Z \otimes Z + \lambda H_x X \otimes X.$$

Meaning the expectation value is given by

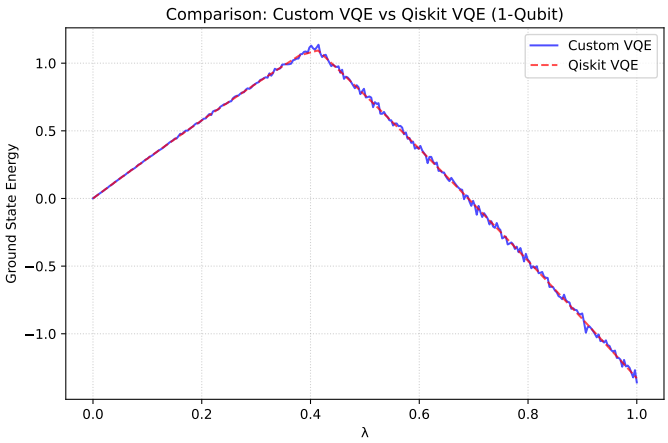


FIG. 9: Comparison between Qiskit and our implementation of VQE. It seems that our implementation is well within a deviance from the benchmark. Here we used 10000 shots.

$$E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle = \sum_i c_i \langle\psi(\theta)|P_i|\psi(\theta)\rangle,$$

where P_i now denotes the two-qubit Pauli strings appearing in the Hamiltonian, and the c_i are the corresponding coefficients.

Another key difference from the one-qubit case is that expectation values now involve two-qubit Pauli operators. These are estimated from joint measurement outcomes in the computational basis. Denoting by n_{ij} the number of times the outcome $ij \in \{00, 01, 10, 11\}$ is observed over N shots, we have for example

$$\langle Z \otimes Z\rangle \approx \frac{n_{00} + n_{11} - n_{01} - n_{10}}{N},$$

$$\langle I \otimes Z\rangle \approx \frac{n_{00} + n_{10} - n_{01} - n_{11}}{N},$$

$$\langle Z \otimes I\rangle \approx \frac{n_{00} + n_{01} - n_{10} - n_{11}}{N}.$$

To measure $X \otimes X$, we rotate both qubits using Hadamard gates before measurement, since

$$(H \otimes H)(X \otimes X)(H \otimes H) = Z \otimes Z.$$

As in the one-qubit case, the total energy is estimated by summing the contributions from each Pauli term, where each expectation value is obtained from repeated measurements. The parameters θ are then optimized using the parameter-shift rule together with the Adam optimizer. In figure 10 we see that the VQE performs better for smaller values of λ, but struggles a little bit for higher values, ecspecially for $\lambda > \frac{1}{2}$. The behavior of the VQE is expected, since a higher λ corresponds to a larger activation of the interacting Hamiltonian. As a consequence, it is naturally harder to approximate the ground state energy. This is due to the H_0 Hamiltonian, being associated with the ground state energies. In 9 we see a similar pattern, where our VQE routine begins to struggle when the interaction strength increases.

Problem (f)

From here, we us the following notation for a Pauli matrix P

$$P_k = I_1 \otimes \cdots \otimes I_{k-1} \otimes P \otimes I_{k+1} \otimes \cdots \otimes I_N.$$

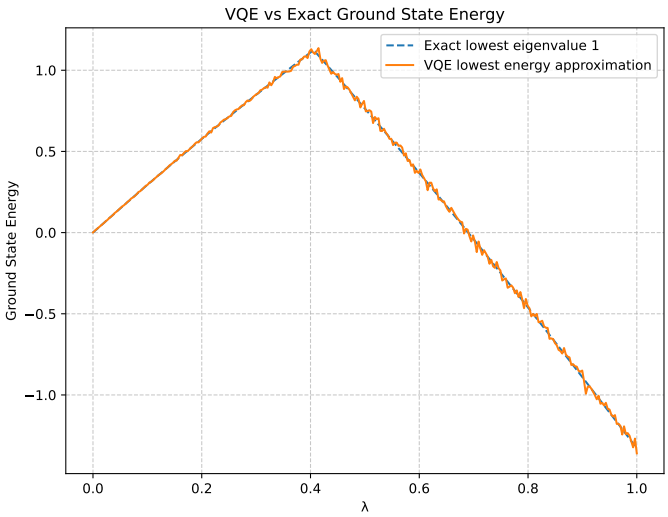


FIG. 10: Comparison between exact diagonalization and our implementation of VQE, using 10000 shots.

Using quasispin, we can rewrite the Hamiltonians as

$$\begin{aligned} H_0 &= \epsilon J_z, \\ H_1 &= \frac{V}{2}(J_+^2 + J_-^2), \\ H_2 &= \frac{W}{2}(-N + J_+J_- + J_-J_+), \end{aligned}$$

with quasispin operators defined as

$$\begin{aligned} J_+ &= \textstyle\sum_p a_{p+}^\dagger a_{p-}, \\ J_- &= \textstyle\sum_p a_{p-}^\dagger a_{p+}, \\ J_z &= \tfrac{1}{2}\textstyle\sum_{p\sigma} \sigma a_{p\sigma}^\dagger a_{p\sigma}, \\ J^2 &= J_+J_- + J_z^2 - J_z. \end{aligned}$$

Our first task at hand is to express $H_{J=1}$ as a matrix in terms of quasispin operators. Each matrix element is computed as $\langle J = 1, J'|H|J = 1, J''\rangle$, where $J' \in \{-1, 0, 1\}$, and J is fixed to be 1. Since $W = 0$, we only get contributions from H_0 and H_1, and for simplification, we calculate then separately.

For H_0 it is easy to see that $\langle 1, J'|H_0|1, J''\rangle = \epsilon\langle 1, J'|J_z|1, J''\rangle = \epsilon J''\langle 1, J'|1, J''\rangle$, since $J_z = J''|1, J''\rangle$. Finally, we can simplify it to $\epsilon J''\delta_{j',j''}$.

Following this procedure we get that

$$H_{0,J=1} = \begin{pmatrix} -\epsilon & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \epsilon \end{pmatrix}.$$

Now we compute $\langle 1, J'|H_1|1, J''\rangle = \langle 1, J'|\frac{V}{2}(J_+^2 + J_-^2)|1, J''\rangle$ with the operators acting like

$$J_\pm^2|1, J''\rangle = J_\pm J_\pm|1, J''\rangle,$$

and

$$J_\pm|J, J''\rangle = \sqrt{J(J+1) - J''(J'' \pm 1)}|J, J''\rangle.$$

Computing this for all elements gives

$$H_{1,J=1} = \begin{pmatrix} 0 & 0 & -V \\ 0 & 0 & 0 \\ -V & 0 & 0 \end{pmatrix},$$

since J_+ and J_- simply excites and de-excites particles it is a matter of counting. Combining these, we get

$$H_{J=1} = H_{0,J=1} + H_{1,J=1} = \begin{pmatrix} -\epsilon & 0 & -V \\ 0 & 0 & 0 \\ -V & 0 & \epsilon \end{pmatrix}.$$

We note that the dimension of the rewritten $H_{J=1}$ matrix cannot be less than 3x3, but when representing it using qubits, the smallest space is 2 qubits, which has dimension 4×4. Meaning that the Hamiltonian must be 4×4, although only the 3×3 dimensional $J = N/2$ symmetric subspace is physically relevant. We start from the individual spin operator definitions

$$j_x^{(n)} = \frac{X_n}{2}, j_y^{(n)} = \frac{Y_n}{2}, j_z^{(n)} = \frac{Z_n}{2}, j_\pm^{(n)} = j_x^{(n)} \pm i j_y^{(n)}.$$

. The factors 1/2 are there since we are working with spin-1/2 particles. Our Hamiltonian, with $W = 0$, is given by $H_{J=1} = \epsilon J_z + \frac{V}{2}(J_+^2 + J_-^2)$. Using the occupation mapping from collective spin operators J to the single-body spin operators given above our Hamiltonian becomes

$$\begin{aligned} H_{J=1} &= \epsilon \sum_n j_z^{(n)} + \frac{V}{2}[(\sum_n j_+^{(n)})^2 + (\sum_n j_-^{(n)})^2] \\ &= \epsilon \sum_n j_z^{(n)} + \frac{V}{2} \sum_{n,m=1} (j_+^{(n)} j_+^{(m)} + j_-^{(n)} j_-^{(m)}) \\ &= \epsilon \sum_n j_z^{(n)} + 2V \sum_{n<m} (j_+^{(n)} j_+^{(m)} + j_-^{(n)} j_-^{(m)}). \end{aligned}$$

By using the definitions of the j operators, and by the fact that we have two qubits, we can rewrite the expression as

$$H_{J=1}^{(2)} = \frac{\epsilon}{2}(Z_1 + Z_2) + \frac{V}{2}(X_1 X_2 - Y_1 Y_2),$$

From the problem statement, we know that the $N = 4$ matris

$$H = \begin{pmatrix} -2\epsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\epsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \epsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\epsilon \end{pmatrix}.$$

Now we want to rewrite the Hamiltonian in terms of Pauli operators. Using the same recipe as for N=2, we rewrite our Hamiltonian as

$$\begin{aligned}
H &= \epsilon J_z + \frac{V}{2}(J_+^2 + J_-^2) \\
&= \frac{\epsilon}{2}\sum_p Z_p + \frac{V}{2}\sum_{p\neq q=1}(X_pX_q - Y_pY_q) \\
&= \frac{\epsilon}{2}\sum_{p=1}^{4} Z_p + \frac{V}{2}\sum_{p<q}(X_pX_q - Y_pY_q) \\
&= \frac{\epsilon}{2}(Z_1 + Z_2 + Z_3 + Z_4) \\
&\quad + \frac{V}{2}\big(X_1X_2 + X_1X_3 + X_1X_4 \\
&\qquad\qquad + X_2X_3 + X_2X_4 + X_3X_4\big) \\
&\quad - \frac{V}{2}\big(Y_1Y_2 + Y_1Y_3 + Y_1Y_4 \\
&\qquad\qquad + Y_2Y_3 + Y_2Y_4 + Y_3Y_4\big).
\end{aligned}$$

Lastly, we rewrite the H_2 term in terms of Pauli matrices. By using the identity $J_\pm = J_x + iJ_y$, we get that

$$\begin{aligned}
H_2 &= \frac{1}{2}W(-N + J_+J_- + J_-J_+) \\
&= \frac{1}{2}W(-N + (J_x + iJ_y)(J_x - iJ_y) + (J_x - iJ_y)(J_x + iJ_y)) \\
&= \frac{1}{2}W(-N + 2(J_x^2 + J_y^2)).
\end{aligned}$$

By the occupation mapping

$$J_x \mapsto \frac{1}{2}\sum_{p=1}^{N} X_p, \quad J_y \mapsto \frac{1}{2}\sum_{p=1}^{N} Y_p,$$

we get that

$$\begin{aligned}
H_2 &= \frac{1}{2}W\left(-N + \frac{1}{4}\left(\sum_{p=1}^{N} X_p\right)^2 + \frac{1}{4}\left(\sum_{p=1}^{N} Y_p\right)^2\right) \\
&= \frac{1}{2}W\left(-N + \frac{1}{2}\left(\sum_{pq} X_pX_q + Y_pY_q\right)\right) \\
&= \frac{1}{2}W\left(-N + \frac{1}{2}\left(\sum_{p=q}(X_pX_q + Y_pY_q) + \sum_{p\neq q}(X_pX_q + Y_pY_q)\right)\right) \\
&= \frac{1}{2}W\left(-N + \frac{1}{2}\sum_p (2I) + \sum_{p\neq q} X_pX_q + Y_pY_q\right) \\
&= \frac{1}{2}W\left(-N + N + \frac{1}{2}\sum_{p\neq q} X_pX_q + Y_pY_q\right) \\
&= \frac{W}{4}\sum_{p\neq q} X_pX_q + Y_pY_q \\
&= \frac{W}{2}\sum_{p<q} X_pX_q + Y_pY_q.
\end{aligned}$$

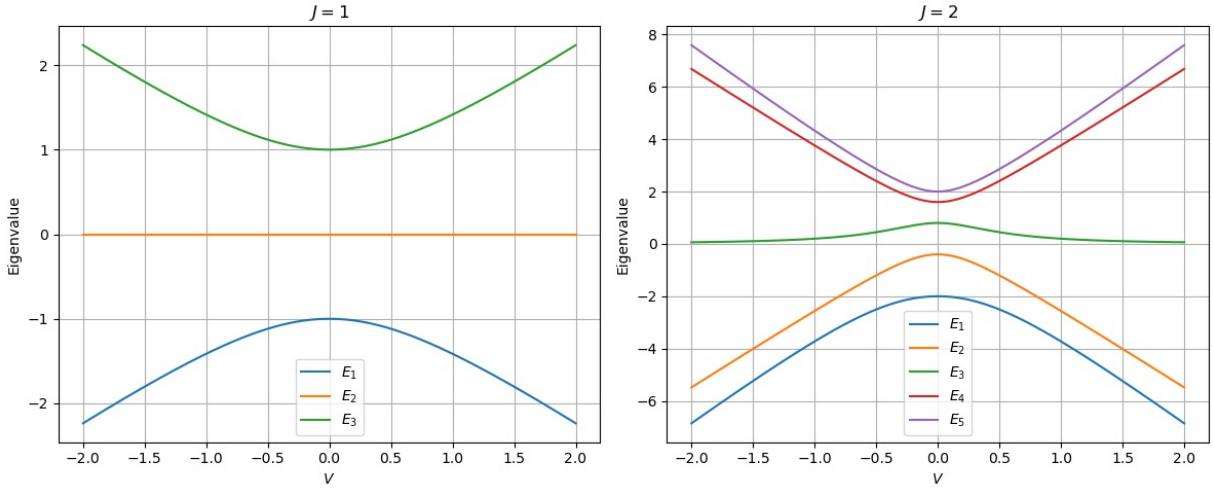


FIG. 11: Plot of all eigenvalues for the $J = 1$, to the left, and $J = 2$, to the right, Hamiltonians with varying coupling strength, $V \in [-2, 2]$. We use $W = 0.2$ for the case of $J = 2$ and $W = 0$ when $J = 1$.

In figure 11 we have included a plot of all the eigenvalues of the $J = 2$ and $J = 1$ hamiltonian, across coupling strength $V \in [-2, 2]$. In both plots we see that the differences between the lower energies and the higher energies diverge as $|V|$ grows.

Problem (g)

To diagonalize more efficiently and in order to reuse the VQE methods for one and two qubits, a block diagonalization of the 5x5 Hamiltonian is performed. Now we get a 3x3 and a 2x2 matrix to be used in the VQE algorithm, and we must rewrite these in terms of Pauli matrices.
After reordering elements of the matrix, we are left with

$$H_{diag} = \left[\begin{array}{ccc|cc} -2\epsilon & \sqrt{6}V & 0 & 0 & 0 \\ \sqrt{6}V & 4W & \sqrt{6}V & 0 & 0 \\ 0 & \sqrt{6}V & 2\epsilon & 0 & 0 \\ \hline 0 & 0 & 0 & -\epsilon + 3W & 3V \\ 0 & 0 & 0 & 3V & \epsilon + 3W \end{array}\right].$$

We start by rewriting the 3x3 block in terms of Pauli matrices, giving us

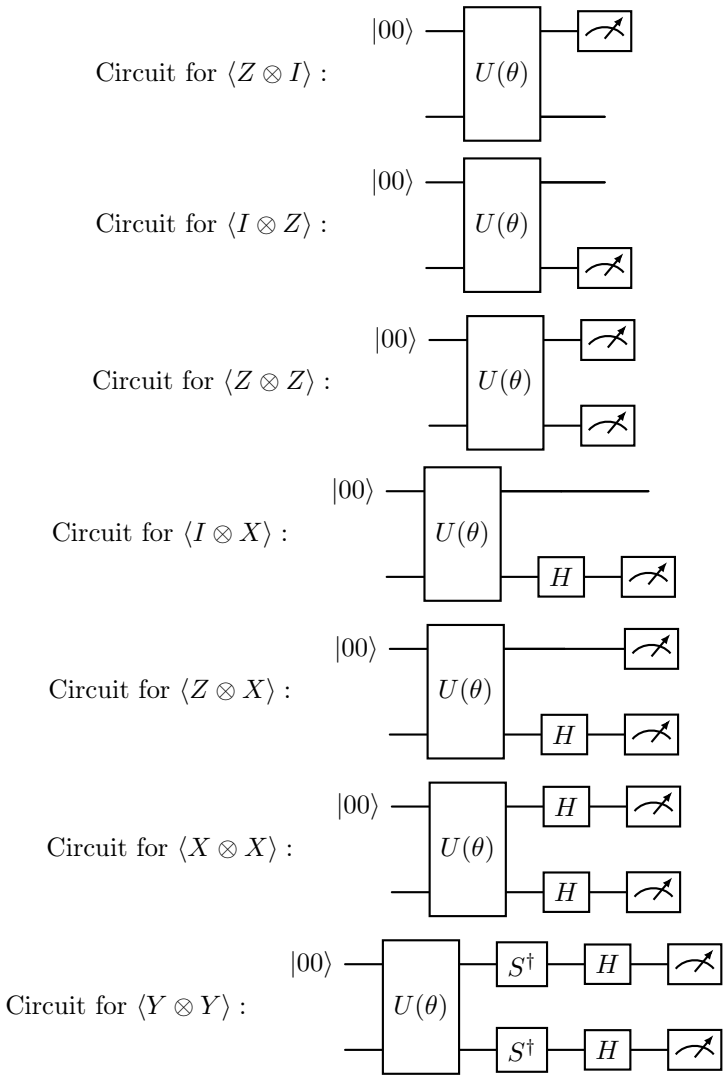
$$H_{3x3} = 2W(I_1 I_2) - \epsilon(Z_1 + Z_1 Z_2) + V\frac{\sqrt{6}}{2}(X_2 + Z_1 X_2 + X_1 X_2 + Y_1 Y_2) - 2W Z_2.$$

This gives a 4×4 two-qubit representation in which the upper-left 3×3 block is the desired Hamiltonian. The 2x2 matrix allows for a simple decomposition, given by

$$H_{2x2} = -\epsilon Z + 3VX + 3WI.$$

In our code, we have been setting up the equations in terms of Pauli Operators, to be able to perform the VQE. We use the same ansatz for the circuit, as we did in problem E, but the circuit must also take the Y terms into account.

Hence we can write a schematic for our circuit as follows, where we omit measurement on the identity.



In figure 12 we set $W = 0.2$ and for this particular plot the shots were 20000. The VQE is capturing the exact energies well for the 2x2 system. However, it starts to struggle for the 3x3 block Hamiltonian, in particular when the coupling strength increases in absolute value. As we saw in problem F, the VQE algorithm seems to struggle when we allow for stronger interactions between particles. This is as expected, since the complexity increases, and the optimization algorithm will find it more difficult to approximate the exact diagonalization. Also, it is worth mentioning that the ADAM method used in VQE could be optimized even further by allowing for model selection, by choosing different hyperparameters. We could also have used more shots or tried different ansatzes. Another possibility would be to add a code-space penalty, to ensure that the VQE stays in the correct subspace. However, as the VQE did not find an energy smaller than the true ground state energy, this did not seem to be the cause of the error in our approximation.

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[1] V. Alsvik, Kortgaard, *FYS5419 - Quantum Computing and Quantum Information*, `https://github.com/AmundAlsvik/FYS5419` (2026), gitHub repository.

[2] D. P. Kingma and J. Ba, *Adam: A method for stochastic optimization* (2017), 1412.6980, URL `https://arxiv.org/abs/1412.6980`.

[3] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2010), 10th ed.

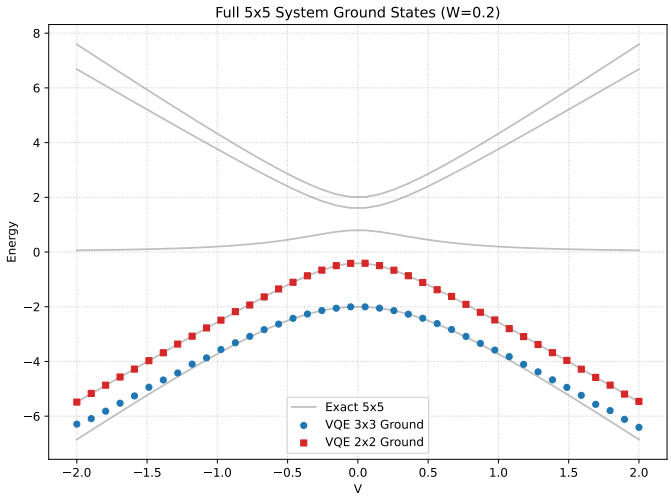

FIG. 12: Comparison between exact diagonalization (line) and VQE (dots) for the ground state energy eigenvalues of the full 5x5 Hamiltonian, using 2x2 -and 3x3 blocks respectively.